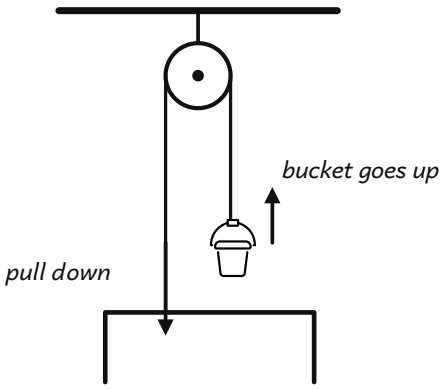
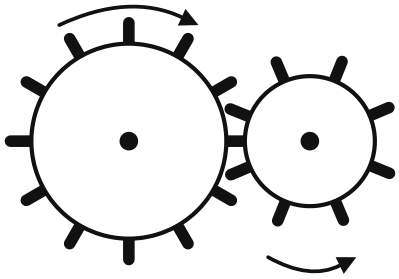
PHYSICAL AND CHEMICAL SCIENCE

# Simple Machines

Levers, ramps, pulleys and gears, and how simple machines make work easier




pulley



gears

**In this pack**

- **Learn it:** Simple machines make work easier ★☆☆
- **Fact check:** Name the machine ★☆☆
- **Think it through:** How do machines help? ★★☆
- **Investigate:** Does a ramp make it easier? ★★☆
- **Real-world case study:** From dheki to rice mill ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 6 Forces and movement (6.3–6.4 Forcemeters).

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · C5 · SIMPLE MACHINES

# Simple machines make work easier

★☆☆ Grade 3

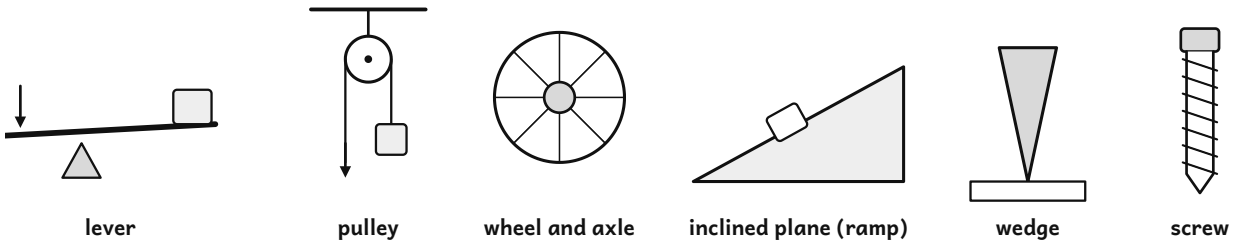
**Name:** **Class:** **Date:**

**Learning goals**

- Explain what a simple machine is.
- Find levers in everyday life and explain how they make work easier.
- Explain how ramps, pulleys and gears help us.

Try to lift the lid off a tin of powdered milk with your fingers. Hard! Now push the end of a spoon under the lid and press down. Easy! The spoon is a **simple machine**.

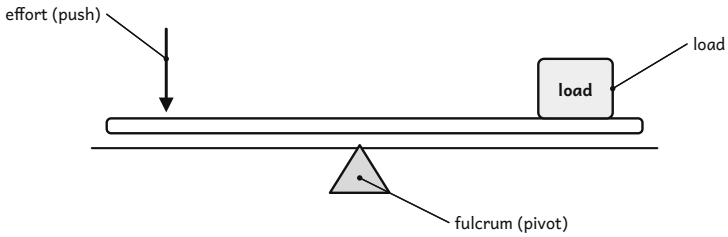
A **simple machine** is a tool with few or no moving parts that makes work **easier**. It may let us use **less force**, **change the direction** of a force, or move something **further or faster**.




*Six simple machines.*

## Levers

A **lever** is a bar that turns on a point called the **fulcrum** (pivot). You push on one part (the **effort**) to move a **load**. The **further** your effort is from the fulcrum, the **less force** you need.

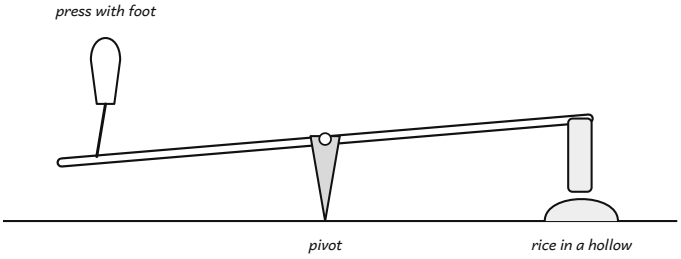



*A see-saw lever.*

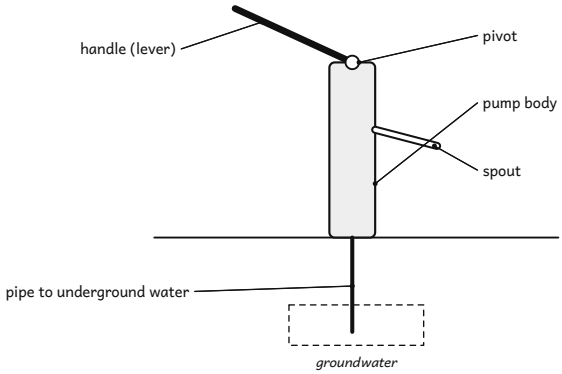
| | | | | |
|---|---|---|---|---|
| scissors | see-saw in a park | spoon opening a lid | wheelbarrow | hammer pulling a nail |

**Bangladesh connection**

A **dheki** is a long wooden lever used in villages to husk rice, and a **tubewell** handle is a lever that lifts water.



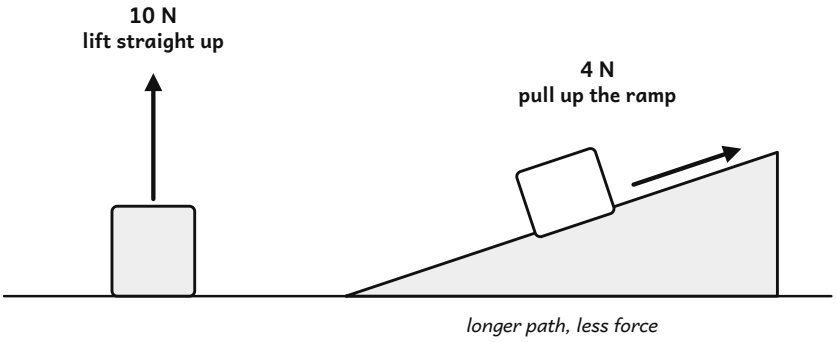

*A dheki.*




*A tubewell.*

## Ramps (inclined planes)

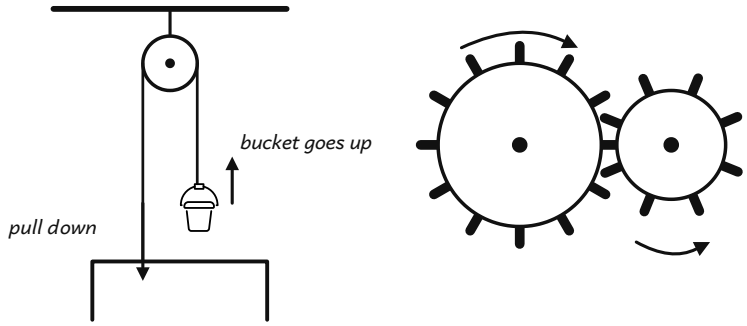
A **ramp** is a slope. It is easier to push a heavy load up a ramp than to lift it straight up: you move it **further**, but with **less force**. Ramps help wheelchair users, people loading trucks, and cars on hill roads that zig-zag up the slope.




*A ramp means less force.*

## Pulleys and gears

A **pulley** is a wheel with a rope over it. Pulling **down** on the rope lifts the load **up**. Pulleys lift buckets from wells, flags up poles and heavy loads on building sites. **Gears** are wheels with teeth that turn each other. Bicycles, clocks, egg beaters and rice mills use gears.




*A pulley and two gears.*

**Key words**

| | |
|---|---|
| **simple machine** | a tool that makes work easier |
| **lever** | a bar that turns on a fulcrum |
| **fulcrum** | the point a lever turns on |
| **effort** | the push or pull you give |
| **load** | the thing being moved |
| **pulley** | a wheel with a rope |
| **gear** | a wheel with teeth |

FACT CHECK · C5 · SIMPLE MACHINES

# Name the machine

★☆☆
Grade 3

**Name:** _____ **Class:** _____ **Date:** _____

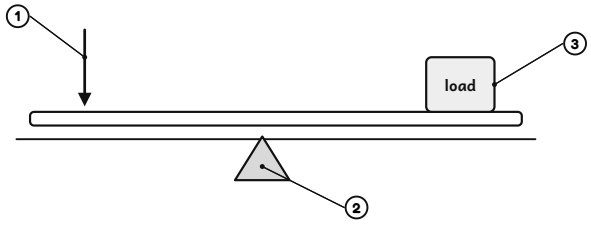
**Learning goals**

- Label a lever.
- Name simple machines.

## 1 Label the lever. [3]

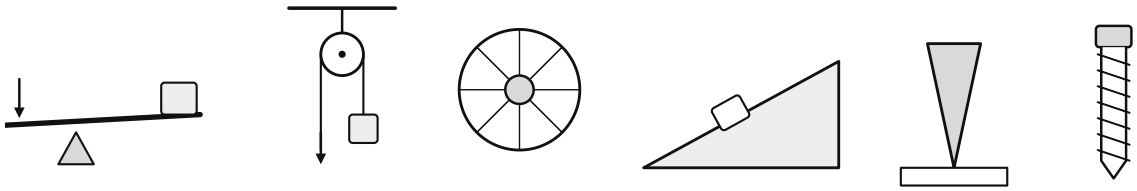
LABELS

**fulcrum (pivot)  load  effort (push)**




1 _____ 2 _____ 3 _____

## 2 Name each simple machine. [6]



A B C D E F

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| | | | | | |

## 3 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| scissors ● | ● **A** lever |
| wheelchair slope ● | ● **B** pulley |
| flag pole rope ● | ● **C** ramp (inclined plane) |
| bicycle chain wheels ● | ● **D** gears |

# How do machines help?



Name: Class: Date:

**Learning goals**

- Explain how machines make work easier.

**1** **Why is it easier to open a tin lid with a long spoon than with a coin?** [2]

**2** **Why do roads up steep hills zig-zag?** [2]

**3** **How does a pulley help someone raise a flag?** [2]

**4** **Draw a lever you use at home. Mark the fulcrum, effort and load.** [3]

My lever

INVESTIGATE · C5 · SIMPLE MACHINES

# Does a ramp make it easier?

★★☆ Grade 3

Name: Class: Date:

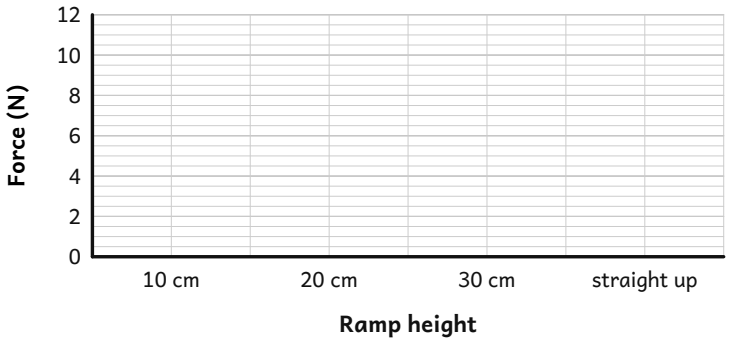
**Learning goals**

- Measure force in newtons.
- Draw a bar chart and conclude.

Class 3C used a forcemeter to pull a toy truck up ramps of different heights. Lifting it straight up needed 10 N.

**1** **Draw a bar chart.** [3]

| Ramp height | 10 cm | 20 cm | 30 cm | straight up |
|---|---|---|---|---|
| Force (N) | 3 | 5 | 7 | 10 |




**2** **Which ramp needed the least force?** [1]

**3** **Complete: the lower (gentler) the ramp, the ___ force is needed.** [1]

**4** **What did they keep the same?** [1]

REAL-WORLD CASE STUDY · C5 · SIMPLE MACHINES

# From dheki to rice mill

★★★ Grade 3

**Name:** **Class:** **Date:**

**Learning goals**

- See how machines change daily life.

Science | History | Work

For hundreds of years, village women in Bangladesh husked rice with *a* **dheki**. One person stepped on the end of *a* long wooden beam. The beam turned on *a* pivot *and* the other end lifted, then thumped down onto the rice to knock off the husks. It was slow, hard work.

Today most rice is husked in **rice mills**. A motor turns **gears** *and* **belts** that work the machines, *and* one mill *can* husk tonnes of rice in *a* day. Some families still use *a* dheki to make special rice flour for winter pitha.

Based on rice husking in Bangladesh.

**1** **What kind of simple machine is *a* dheki?** [1]

**2** **How did the dheki make work easier?** [2]

**3** **Write one good thing *and* one problem *about* rice mills.** [2]

# Show what you know

★★★
Grade 3

Name: Class: Date:

**Learning goals**

- Use machine words.

## 1 Find the words.

| N | G | M | L | E | V | E | R | H |
|---|---|---|---|---|---|---|---|---|
| L | E | A | C | H | L | Y | G | H |
| P | A | E | F | F | O | R | T | T |
| D | R | I | R | A | K | P | K | V |
| P | U | L | L | E | Y | H | N | F |
| K | W | W | H | E | E | L | R | G |
| L | O | A | D | W | Y | R | A | F |
| D | Y | G | P | S | V | L | U | S |
| H | F | F | B | V | R | A | M | P |

**effort gear lever load pulley ramp wheel**

## 2 The point a lever turns on is the... [1]

- ◯ load
- ◯ fulcrum
- ◯ gear

## 3 A ramp lets you use... [1]

- ◯ more force
- ◯ less force
- ◯ no force

## 4 Challenge. Design a machine to lift a bucket of sand to a first-floor window. Label the simple machines you used. [3]

My machine

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain what a simple machine is | ○ | ○ | ○ |
| find the fulcrum, effort and load | ○ | ○ | ○ |
| explain how a ramp helps | ○ | ○ | ○ |
| explain how pulleys and gears help | ○ | ○ | ○ |
| measure force on a ramp | ○ | ○ | ○ |